Equipment

ONR FY2027 YIP, NOFO N0001426SF004. PI: Sterling G. Baird, Brigham Young University

Equipment available. The equipment below is in place and used at no cost to the project except where a facility charges hourly rates (shown in the budget).

- AMAZEMET rePOWDER ultrasonic atomizer with induction crucible module, to 1,300 °C, with the manufacturer's coated plates and boron nitride nozzles for copper alloys (PI laboratory, BYU; bought with start-up funds).
- Liquid-handling gantry (CubXL, Ursa Laboratories) with a pipette, a vial capper, and cameras, run by scripts (PI laboratory); it runs the corrosion screen and samples immersion solutions.
- Powder handling, sieving, and specimen preparation equipment in the PI laboratory [TODO Sterling: list the main items; add a Hall flowmeter funnel if one must be bought].
- SEM, EDS, EBSD, and TEM at the BYU Microscopy Center (Apreo C, Helios 600, Verios; Tecnai TF-20).
- Agilent 8900 ICP-MS/MS and XPS (BYU Spectroscopic Facility); X-ray diffraction (BYU XRD Facility).
- Metallographic polishers, a Keyence VHX-7000 digital microscope, and CNC machining (BYU).
- Tensile testing: [TODO Sterling: BYU load frame or test service].
- BYU high-performance computing cluster.
- Blown-powder laser DED system at [TODO Sterling: facility; candidate: University of Utah, Prof. Zak Fang], used through purchased sessions [TODO Sterling: system model; likely an InssTek MX-Lab].

Equipment to be bought with project funds. One potentiostat with corrosion software and a flat cell, Year 1, estimated at $18,000 [TODO Sterling: attach the vendor quote, as the NOFO requires]. The electrochemical corrosion screen is one step of the workflow that the search method schedules, so the instrument must be available every day and run by the laboratory's own scripts on the gantry above. No equipment that can serve this role is available to the project at BYU [TODO Sterling: decide: a module potentiostat with impedance on the gantry (about $600 in the lab's survey) plus a borrowed laboratory instrument for cross-checks would remove this item and most of the budget line; see cubxl-electrochemistry.md].

Government-owned equipment. None.